

Lesson 02 · Functions, lines and market equilibrium

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If $f(x) = 3x + 15$, evaluate $f(2)$.

☐ A 6

☐ B 17

☐ C 21

☐ D 36

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Q2 · Rank 1 / Foundation

If $f(x) = 3x + 15$, evaluate $f(0)$.

☐ A 0

☐ B 3

☐ C 18

☐ D 15

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Q3 · Rank 1 / Foundation

For the supply function $P = \frac{1}{3}Q + 7$, determine P when $Q = 12$.

☐ A 11☐ B 4☐ C 43☐ D 19

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Q4 · Rank 2 / Core

State the value of the slope and y intercept for the line $y = 5x + 9$.

☐ A Slope 9; intercept 5☐ B Slope 5; intercept 9☐ C Slope -5; intercept 9☐ D Slope 5; intercept -9

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Q5 · Rank 2 / Core

For the line $3x - 5y = 8$, find the value of x when $y = 2$.

☐ A $-\frac{2}{3}$

☐ B $\frac{8}{3}$

☐ C 6

☐ D 18

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Q6 · Rank 2 / Core

A taxi firm charges a fixed cost of USD 4 plus a charge of USD 2.50 a mile. Write down a formula for the cost, C , of a journey of x miles.

☐ A $C = 6.5x$

☐ B $C = 4x + 2.5$

☐ C $C = 2.5x$

☐ D $C = 4 + 2.5x$

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Q7 · Rank 3 / Application

An airline charges USD 300 for a flight of 2000 km and USD 700 for a flight of 4000 km. Assuming a linear model, estimate the cost of a flight of 3200 km.

☐ A 480☐ B 640☐ C 540☐ D 500

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Q8 · Rank 3 / Application

A taxi firm charges a fixed cost of USD 4 plus a charge of USD 2.50 a mile. Hence, or otherwise, work out the distance of a journey which costs USD 24.

☐ A 8 miles☐ B 9.6 miles☐ C 10 miles☐ D 11.2 miles

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Q9 · Rank 3 / Application

If the supply function is $Q = 4P - 25$, determine the lowest possible price that could be charged.

☐ A 25☐ B 0☐ C $\frac{25}{4}$ ☐ D $\frac{4}{25}$

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Q10 · Rank 4 / Challenge · OPTIONAL

The demand and supply functions of a good are given by $P = -3Q_D + 75$ and $P = 2Q_S + 15$. Determine the equilibrium price.

☐ A 39☐ B 12☐ C 60☐ D 15

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